

Computational Notebook as a Tool for Supporting Black Engineers Non-Profit Organizations in the Context of Workforce Participation

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ABSTRACT

Data is frequently referred to as the ‘new oil’ due to its high strategic value and the possibilities it brings. Additionally, its use by powerful organizations has the potential to help, but also to cause harm and function as an additional oppressive mechanism towards vulnerable communities. Hence, there is a need for providing easier methods for people and organizations to access, analyze, and interpret reliable, publicly available data (e.g., U.S. Census), and create data narratives that reflect their social position in the society and inform how to improve that position. In this case study, we document our journey on using computational notebooks to increase autonomy for organizations tackling racial and social justice issues by making meaningful use of publicly available data. The project considered a participatory action research approach run in partnership with a non-profit organization supporting Black engineers. The reported case study involved participants representing staff and volunteer leadership of the organization. We expect that the presented reflections help move towards a more equitable way of performing data analysis while using popular and freely available technologies.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

Participatory Action Research, Computational Notebooks, Jupyter Notebook, Data Action, Social Justice, Racial Justice.

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1 INTRODUCTION

Machine Learning, Artificial Intelligence (AI), and User Experience (UX) frameworks and technologies have been applied in multiple projects aiming at social [17, 33, 51] and racial justice [8, 29, 55]. All of these technologies have been critiqued for accelerating, embedding, and exacerbating harms experienced by marginalized communities [3, 4]. Addressing these issues is at the forefront of what the civil technology, data justice, and AI fairness communities prioritize and advocate against. In order for this sociotechnical work to be effective, engagement of the communities that are harmed is necessary and provides an opportunity for social justice non-profit organizations (SJNPs) to offer a ground truth understanding of the experiences of their constituents. According to Williams [54], “*data can be used as evidence that can help or harm*,” hence, supporting organizations whose mission it is to perform their own data analysis and advance storytelling is key to foment alternative points of view not only in the creation, but also during the use, iteration, and repurposing of technologies.

In this case study we detail the use of computational notebooks to support rapid data analysis, trend monitoring, and short-term prediction of United States (US) workforce participation for Black engineering professionals. Our partner organization joined the project in late 2021 and began participation in 2022 as a part of an initiative aiming at developing technology solutions for social and racial justice, Tech for Justice. The Tech for Justice research initiative was launched in 2020 to promote collaboration with communities and non-profit organizations to develop technical solutions that support and enable greater social justice for historically marginalized groups. To ground the specific research projects that would be a part of Tech for Justice, researchers engaged with non-profit organizations in discovery conversations about the technical challenges and issues that were pertinent to their organization’s mission and work. This project was developed in collaboration with staff members and volunteer leadership of the National Society of Black Engineers (NSBE) ¹, a 501(c)(3) membership nonprofit organization based in the US. In discovery discussions, the problem space of workforce

¹<https://www.nsbe.org/>

participation, labor outcomes, and educational attainment for the intersectional populations that the organization advocates for was identified. Through subsequent conversations with other possible partners, the topic of workforce participation was broadened to include various occupations and further highlighted the need for ready data analysis of this information for multiple organizations with similar missions. In the process of uncovering this data need, we found that there are limited to no ready-to-use data visualization tools publicly available that represent the US workforce. Much less, tools that enable delineation across race, gender, occupation, education, compensation, and other variables wherein issues of racial and social justice are best understood with data. We realized in conversation with our partners how frequently this data is reported and updated, further necessitating the ability for organizations to rapidly complete analysis on their own with as little complication as possible. For approximately a year, a series of bi-weekly collaborative partnership conversations took place and included information sharing, simple user testing, feedback gathering, and stakeholder assessment to inform the two versions presented in this case study. Often, interim data products were shared to support timely organization activities, which also honed the product iterations and further aligned prototype development for advancing the advocacy and programmatic efforts of NSBE.

In the US, the federal government collects data from many sources and programs and, in turn, makes this data available as a valuable public data source. Increasing technology capabilities within the federal government, the federal laws related to public records, and the will of political administrations to make open data access a reality has led to various public repositories across agencies available to the general public. Some government agencies have greater investment in managing data and hosting infrastructure to reduce the technical workload on both the curious public and serious researchers attempting to access and make sense of data. Others, however, focus their data investments in data collection and place less emphasis on data analysis and tooling. This then requires data practitioners to have significant data science, management, and handling skills to use the data meaningfully whether through massive data tables or APIs. This creates a gap in access and use that our case study seeks to address.

Valid critiques of government data resources include the issues of under-counting and exclusion. This is especially true for marginalized people and those discouraged from public and political participation for various reasons, including personal and geographic, and those left out due to the methods of data collection [7, 36, 53]. Authors like D'Ignazio and Klein in *Data Feminism* [15] and Williams in *Data Action* [54] offer several case studies to highlight how data structures, systems, and collection methods can be oppressive, harmful, or exclude some groups. These authors also suggest approaches to redesign data systems to be more inclusive and to serve positive outcomes for more groups. Their critique, and that of others sharing a nuanced, social justice-oriented lens, highlight the concerns with public and traditional data sources themselves. The limited and narrow inclusion of all groups in public big data is a main area of concern when leveraging these data in the context of a social justice organization.

The second significant limitation is the ability to make practical use of open data. Generating actionable insights from any data is a

complex task, especially given the multiple sources, codes, formats, and workflows applicable for open government data. Data analysis becomes complicated when using US federal data as different agencies have their own strategies and approaches to data management, which can also shift regularly between administrations. Some agencies provide enhanced data tools, such as the US Census Data Explorer², however such an investment in data transparency is not available through all agencies. The complexity of the task of data analysis requires developer-level skills when large tables of raw data are the source. This hurdle can quickly become insurmountable for SJNPs, which rarely if ever have data scientists on staff or ready access to the necessary data tools to support analysis. The task of developing data visualizations by leveraging public or proprietary data is often a small part of a single staff member's responsibilities. These data practitioners or investigators have the skills to understand and create data reports with a set of tools to create charts and visualizations for a public that may have basic data literacy skills. This set of opportunities: government data complexity, issues of inclusion, the need for a social justice lens, and shifting data strategies and tools motivate this initiative, the Data Hub and Accelerator project.

Exploratory programming has become a promising approach to enable the transition from consumers of descriptive statistics to creators of data narratives and insights. It is also becoming popular with the support of literate programming tools [28]. The most common form of such a tool is computational notebook and the most popular notebook nowadays is Jupyter [37]. Computational notebooks support analysis and the creation of narratives around computing and data analysis, exposing rational and data analyst's line of thought [28, 37]. Although criticisms exist around poor software engineering practices allowed by computational notebooks, they are also transforming how programming languages and data analysis are taught. This is accomplished through bringing new learning possibilities as first lines of code, visualizations, documentation, and machine learning models can be created faster than through previous options [6, 9, 22, 27, 37, 44, 48].

In terms of adoption of technologies, the literature also shows that the first meaningful experience is important for attenuating symptoms of technophobia and/or computer anxiety for technology newcomers [19]. In the context of social and racial justice, this aspect is permeated by additional factors such as the skills needed to use the technology and skepticism of using empirical data to support social justice efforts [10, 18].

In this context, the present work starts by questioning the need for a new technology in the first place [45] and builds on top of Participatory Action Research (PAR) so that created artifacts (if any) are meaningful and transformative for people and/or organizations acting in the realm of social and racial justice. PAR aims at involving the participation of those people experiencing issues, who take action to produce emancipatory social change [11]. The work on PAR connects with the theory proposed by Paulo Freire [54]. According to Freire [20], education is a powerful way to help liberate the oppressed by empowering people to (1) actively engage in conversations with those in power and (2) construct their own narratives. By involving the people most impacted by technology,

²<https://data.census.gov/>

the Design from the Margins (DFM) approach emerged. DFM can be defined as “a design process that centers the most impacted and marginalized users from ideation to production, in order to expand the scope of user needs, experiences, and risks considered during the development of technologies.” [43]. More recently, the Data Action method has been proposed as a way to consult domain experts on data analysis, building data analysis tools, promoting participation of impacted communities, repurposing (or hacking) technologies, and sharing assets developed. Hence, this work connects PAR and DFM by involving people impacted by existing issues related to participation in the mainstream US labor force. Moreover, this work is also inspired by the Data Action method to promote the use of computational notebooks as empowering technology so non-profit organizations can create their own data narratives while supporting their communities in job and career matters. In sum, our work follows “the idea that engagement with data is one way to help overcome the oppression it might cause” [54].

1.1 Contribution Statement

This work contributes with a case study, part of a project called Data Hub and Accelerator, in which PAR was applied to empower an organization supporting Black engineers in performing their own data analysis. The project counts on iterations of a data analysis asset and the most recent version leverages good attributes of computational notebooks in terms of ease of use, data analysis, and visualizations. According to Williams [54], visualizations carry bias from their creators and, at the same time, “visualizations have been effective tools for swaying the public because they retain the legitimacy of data.” Hence, it is of key importance to allow organizations and communities to create their own narratives, data analyses, and visualizations through the process of data literacy and activism.

The remainder of this case study is structured as follows: the next section (section 2) presents the related work, section 3 details the project and the two prototypes studied, section 4 dives into the method used to evaluate the current prototype version, section 5 discusses outcomes obtained so far, and section 6 presents our conclusions and next steps of this research.

2 RELATED WORK

This section overviews research around human-AI interaction for decision making, participatory research in the context of social and racial justice, and the use of computational notebooks in different multi-stakeholder contexts.

When considering human-AI interaction, full automation is sometimes not desirable in high-stakes domains such as criminal justice and healthcare, but these domains also bring attention to the concept that full manual approaches may be inaccurate and time-consuming [30]. This in turn leads to the increasing interest in human-AI decision making. The publication of scientific papers around human-AI decision making and human-AI interaction tripled in the last couple of years [30], while doubling times of science take more than 17 years [5]. Shneiderman recently proposed a two-dimensional Human-Centered AI (HCAI) framework that shows possibilities of combining human control and automation [47]. Green et al. [24] contrasts how sometimes decision making is

done with a human supervising an algorithm’s task (human-in-the-loop) with situations when an algorithm participates in a human task (algorithm-in-the-loop) and how the former has its own characteristics. Human-AI partnership for decision making is being used in multiple contexts such as recidivism [23, 25, 52], finance [24], healthcare [33], education [34], job promotion [2], environment [42], combination of multiple AI outcomes [13], and interactive explanations and visualizations for fairness/explainability [16, 32, 50]. Like many sociotechnical problems, all of these contexts for AI tool use receive fair criticism both from societal and technical perspectives. Having in mind participatory research in the context of social and racial justice, literature presents works around environmental justice [12, 21, 35, 39, 46], social justice [14, 26], equal access to wealth [13], and how social justice should move from colonial to more participatory practices [40].

Finally, literature also presents multiple approaches to using computational notebooks in different settings and with different goals as a multi-stakeholder approach. This approach can be seen in the context of health care [1], data storytelling assistants [31], analyzing sensemaking over time [41], hobbyist application in the creation of physical artifacts [49], and frameworks for improving reuse and history management of notebooks’ cells [9]. Computational notebooks are popular, easy to setup, and support the creation of computational narratives and visualizations. However, due to their flexibility, they also lack features for enforcing software engineering good practices that would be expected in an IDE (Integrated Development Environment) [9, 37]. Hence, this work proposes a way of using computational notebooks to aid decision-making in the context of non-profit organizations that support social and racial justice. More specifically, using computational notebooks to facilitate non-profits’ ability to monitor workforce participation and create narratives, enhanced with relevant data, representing the experiences of their constituents.

3 THE DATA HUB PROJECT

This section details the Data Hub project, how PAR was applied, our interactions with stakeholders, and how stakeholder knowledge was fundamental to creating a prototype that may be seen as simple in terms of technologies used, but consolidates data knowledge and understanding of needs in terms of publicly available data analysis for supporting data-grounded actions. By having in mind the PAR principles of (1) authority of direct experience, (2) knowledge in action, (3) research as a transformative process, and (4) collaboration through dialogue [11], the Data Hub project values NSBE authority and direct experience on data needs from the community they support and their know how about actions that change people’s lives. In addition, the research project aimed to be transformative in terms of data analysis, and by having continuous interactions with stakeholders.

3.1 First prototype

The development of Data Hub, a data visualization tool for labor force statistics, began with an exploration of existing dashboards and an evaluation of development platforms. By researching current labor force statistics dashboards, valuable insights into design principles and functionalities were gained. In turn, R Shiny stood

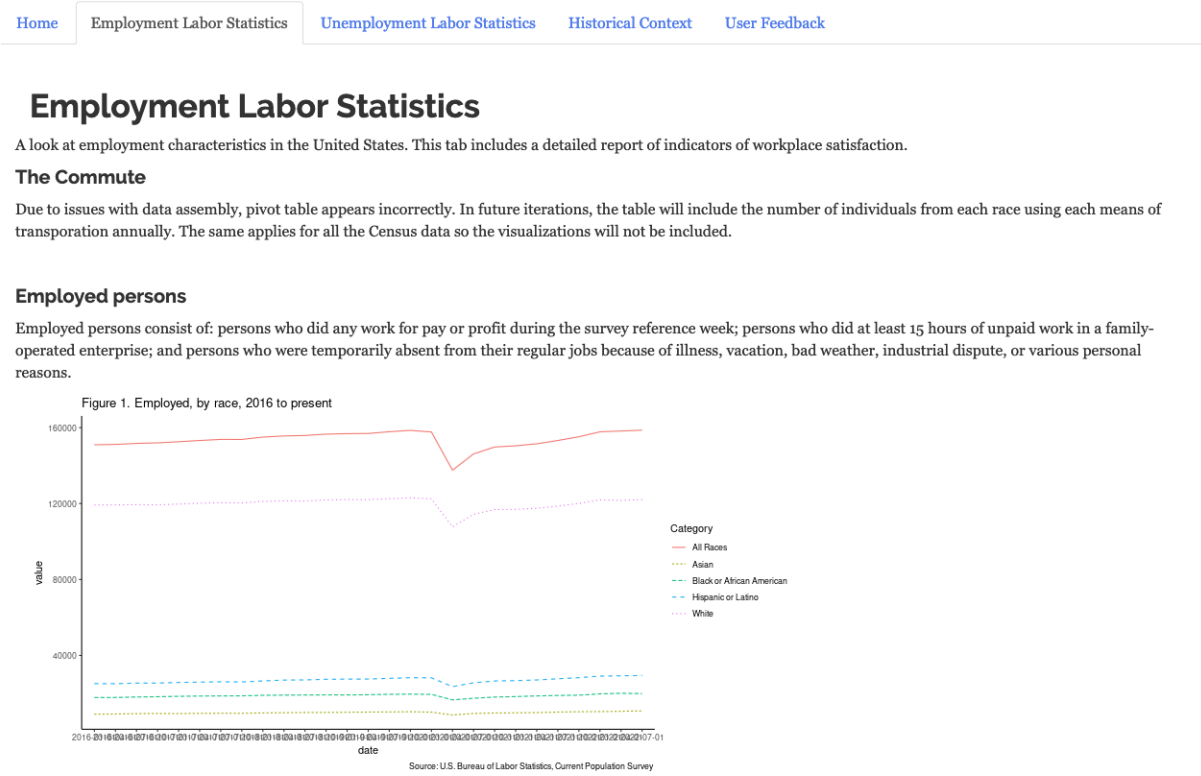


Figure 1: Screenshot of the first version of the prototype.

out as a promising platform due to its reputation for leveraging US Census and Bureau of Labor Statistics (BLS) data, as highlighted by Perkins [38]. The decision to utilize R Shiny was further solidified based on its reproducibility, open-source nature, and the availability of extensive troubleshooting resources. These factors made it an ideal choice for creating the initial prototype (Figure 1).

Despite the advantages of using R Shiny, challenges arose when integrating it with the cloud provider of our company. Scalability and compatibility issues also emerged, necessitating the exploration of solutions that would enable seamless integration and the consideration of alternative design approaches that aligned with cloud technologies. Discussions with our partners raised the need to enable use of multiple data streams in place of a static set of CSV (comma separated values) files and this became a concerning limitation for the application. In turn, this was further exacerbated by the frequency with which the BLS data is updated, some as frequently as weekly. Requiring partners to search out, download, and upload new CSV files in order to have up to date data was not within our planned scope.

The Data Hub prototype, based on an analysis of current labor statistics dashboards, incorporated various design trends to create a minimum viable product (MVP) that integrated longitudinal trends of labor force statistics, data summaries, and data source transparency statements. Moreover, the prototype introduced unique features not found in existing dashboards, including historical context of data findings, user feedback components, and the ability

to explore desegregated data based on race, age, and sex. These features were identified as important for enhancing user experience and providing a comprehensive understanding of labor force statistics in conversations with our partners.

Following the development of the Data Hub MVP, three external stakeholders were engaged in a focus group session to gather feedback. Valuable insights were obtained from the stakeholders, including the need to compare labor force statistics to previous periods of economic recession and the importance of using employment and unemployment indicators. To address this feedback and cater to the requirements of external stakeholders, a narrative synthesis was incorporated. The synthesis provided a detailed description of the data and compared it to previous economic recessions. In addition, gray shaded areas were added to indicate periods when data trends aligned with previous recessions. Addressing these comments required the integration of multiple tables from the US Bureau of Labor Statistics. This posed the challenge of integrating an SQL database, R Shiny, and the cloud platform adopted, resulting in slower load speeds of the web application.

In addition to the data representations themselves, the stakeholders identified the need to review specific occupational categories. Through conversations with partners, we identified a total of 27 engineering and other STEM (Science, Technology, Engineering, and Mathematics) occupations to include in the Data Hub. The organization also expressed interest in gauging additional information about STEM adjacent occupations. However, upon mutual

review of those data identified, it was decided that these occupational categories, while important, would not provide clear enough information about the populations of interest. These interviews also raised questions about the engineering workforce that the research team had not considered before. Industry of employment, career progression, management representation, and second jobs were raised as topics that the organization staff and leadership were interested in monitoring based on their understanding of how the organization's constituents participate in the US STEM workforce.

This initial version of the tool and evaluation with partners offered important lessons about database, application, and cloud platform selection. Specifically, improvements were needed to enhance the load speed of the web application and streamline the process of effectively calling code from various sources. Providing the opportunity to limit the number of graphs displayed on a single page through an interactive visualization was another important feature for our team to take into consideration as the first version provided two pages containing 15 to 17 graphs and supporting text. These considerations are vital to ensure the seamless integration of the Data Hub tool with the desired technologies. By overcoming these hurdles, Data Hub can be better equipped to meet the needs of external stakeholders and provide an efficient platform for analyzing labor force statistics.

3.2 Second prototype

The requirements for the second version followed the lessons learned in the previous version. These can be broken down into three main categories. First, using a computational infrastructure available on the cloud to ease data analysis so that partners would not have to configure the computing environment themselves. Second, creating an easy to download computational notebook in case partners want to perform data analysis locally. Third, ensuring the use of a computational notebook with an interactive data analysis tool available on the web. Given these lessons, the second prototype uses a cloud infrastructure (IBM Cloud Pak for Data) and the computational notebook selected counts on an interactive data analysis feature in Python, using the PyGWalker library³. The PyGWalker library aims at simplifying data analysis on notebooks through interactive visualizations inspired by Tableau software⁴.

The computational notebook was developed in Python and is publicly available at: <https://ibm.biz/datahub> (Figure 2). Python was chosen as the language for the second version of the prototype in case partners want to iterate on the computational notebook provided and because Python is the current trending language⁵. Python's popularity can be attributed to multiple reasons including its ease of use, open-source nature, and powerful capabilities⁶. The computational notebook consists of data analysis and data visualization tools for the "Engineering data" and "Median hourly earnings" datasets. These two datasets were selected from the first prototype. The data used in this version of Data Hub was sourced from the Department of Labor, Bureau of Labor Statistics (BLS)⁷, where the "Engineering data" dataset combines 50 data series from

the BLS and the "Median hourly earnings" dataset combines 12 data series from the BLS.

4 METHOD

This section details the methods used in the analysis of the second version of the prototype. The overall goal of this study is to assess whether the proposal of giving partners access to and different ways of creating a narrative (i.e., the computational notebook shared with them) is helpful for their activities, what features were necessary to develop their data representations, and what supports were needed to ensure a high adoption and use rate for the tool.

Materials considered in the study include the computational notebook provided, the data analysis provided, and a questionnaire to gather participants' feedback.

In our interviews and guided exploration exercises with partners, we aimed to determine additional gaps in usability, identify additional data to include in the platform, and opportunities for employing computational notebooks in the context of supporting organizations like NSBE. To note, interviews and guided exploration were conducted using our second version of the platform.

The overall goal of the interview sessions was to assess whether giving interviewees access to Data Hub and creative authority over how to structure their individual narratives (i.e., the notebook we shared with them) would be helpful. In this context, a study was performed considering how partners use and repurpose (if at all) the provided data analysis computational notebook. To this end, the computational notebook was provided with proper guidance (textual information and a demo video) on how to use it. Two weeks later, our team performed remote interviews using a video conferencing system. The rationale for creating a two-week buffer was to give participants time to interact with Data Hub such that they could have a meaningful first experience. The interview was semi-structured and covered topics around experiences with the computational notebook, situations and tasks in which they used Data Hub, if they repurposed the computational notebook, narratives created, and any feedback they would like to provide to the team.

In addition to one staff member who was consistently engaged in the project, another staff member, a likely user of the Data Hub from a different functional area, and three volunteer leaders were recruited to participate. A "launch" meeting took place to introduce the study and its goals. This meeting was followed by a guided study sheet, sent via email, that provided step-by-step instructions on how to access and use the Data Hub, as well as two demo videos showing how to interact with the notebook. Due to existing time demands for our participants, we offered three modes of participation: live recorded video conference with a research team member, self-recorded screen capture video completing the tasks, or completion of the tasks with screenshots and feedback notes. We realize that this might cause some inconsistency, however time resources to take on additional tasks like these are limited in mission-driven non-profit organizations. This being said, we did seek feedback from additional leaders and look forward to engaging their experience with the tool and findings.

³<https://github.com/Kanaries/pygwalker>

⁴<https://www.tableau.com>

⁵<https://octoverse.github.com/2022/top-programming-languages>

⁶<https://github.blog/2023-03-02-why-python-keeps-growing-explained/>

⁷<https://www.bls.gov>

Data Hub

Interactive visualization for labor data.

Autor da versão
Vagner Figueredo de
SantanaLinguagem
Python 3.10Última atualização
4 de jan. de 2024,
11:03

Labor data analysis

In [19]: # The code was removed by Watson Studio for sharing.

In [20]: # The code was removed by Watson Studio for sharing.

Engineering data

Data Visualization

Chart 1 + New

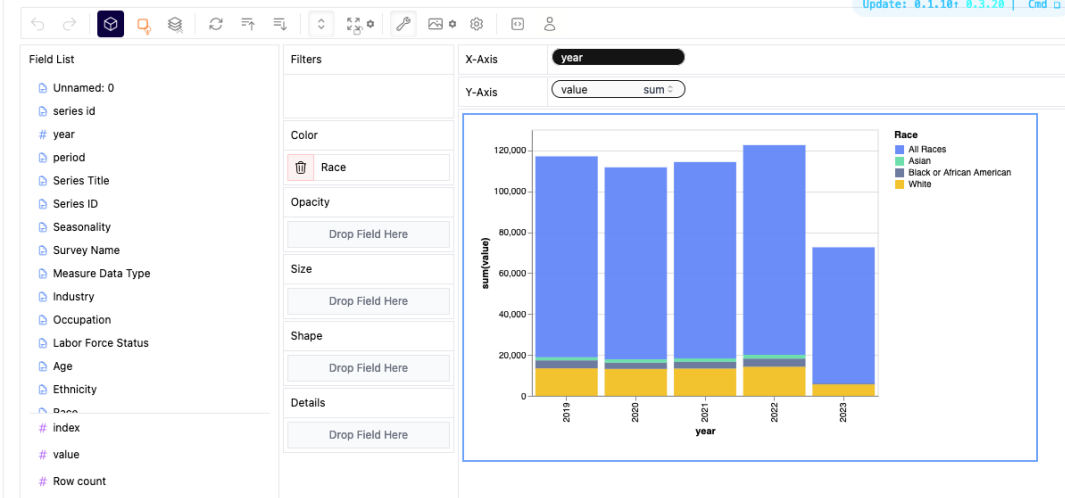


Figure 2: Screenshot of the second version of the prototype.

The tasks we asked participants to perform are as follows:

- (1) Identify trend(s) (if any) in terms of Black or African American employment in engineering.
- (2) Identify trend(s) (if any) in terms of median hourly earnings for Black or African American men vs. Black or African American women.
- (3) Based on one of the previous analyses, create a filter and select a specific series title for your analysis.
- (4) Identify a numeric field and set it as a size attribute for a circle plot.
- (5) Identify a categorical field and set it as the color attribute for a line plot.
- (6) Create a filter for the field 'measure data type' that you believe is more adequate.

After completing the given activities, participants were asked five feedback questions to explore their experience, share the data outputs of each of the tasks, and ask additional questions that were raised for them while using the tool. There were three categories of feedback from users regarding their experience with the Data Hub activities and tool interaction: user interface, data meaning/contextualization, and actionable data.

5 RESULTS

The user interface, primarily as designed and defined by PyGWalker, left a bit to be desired for users. In written feedback as well as in conversation, understanding and ability to use the toolbars in PyGWalker was not intuitive, nor did the tool provide hints. The user guide that we provided did not include a description of each toolbar icon, so navigating this became frustrating for users. There were also concerns about data meaning and how it is communicated through the user interface. Since users can change their view of the data by interacting with the notebook, changing variables and parameters, there was a built-in expectation that the graph would provide details on how the data was measured and update labels appropriately. Similarly, users wanted to see the values along the graph itself, as well as captured somewhere with the graph created. In an introductory conversation with a participant, the difference between the static graphs in the R Shiny MVP and the fully interactive PyGWalker version was called out; the user specifically asked for some graphs to just be readily available without the requirement of creating a graph on their own. As it relates to data meaning, some intersectional populations were difficult to identify in the datasets and graphs. The process to identify these populations was not intuitive in the tool and some frustrations with the

limitations of the data were discussed. For example, one user mentioned that ethnicity was simply a binary in this dataset. While we discussed how this is a reflection of Hispanic/Latino or not and a built-in systemic limitation of the BLS data collection process, this offered an example of why even a collaboratively designed data tool will still be limited by systemic data practices. One of the main motivations for NSBE in seeking this tool was to create actionable data for the organization's programs and missions. In a standing meeting with two of the partners, how to make this data actionable by the organization was discussed. Some of the data included in the Data Hub is rather static, updated only annually, while other data with less disaggregation ability is updated more frequently. It was mentioned that data tables updated more frequently do not distill race, gender, or specific occupation, thereby making it more difficult to explain with data the observed experiences of the organization's members and constituents. Enhancing these data gaps with organizational data to make a visualization more powerful and support a specific programmatic/policy action was raised as a solution.

Perceived benefits of the computational notebook provided include identification of trends and filters supporting the identification of nuances. A participant reported:

"Trends tend to move in same pattern between groups. Filter by occupation would allow to see the nuances between groups and show occupation where there is the greatest gap."

While another participant reflected on trends and actions taken by organizations:

"The trends point to engineers in the working industry steadily increase after the pandemic yet, there was a massive drop in 2023. [...] What are colleges/companies doing to prepare diverse talent?"

Another participant reflected and reported multiple questions after exploring trends:

"What specific factors contributed to the drastic reduction in the number of Blacks entering engineering occupations in 2023? Are there industry-specific challenges or systemic barriers that disproportionately affect Black individuals in the engineering workforce during economic downturns? How can targeted interventions and support programs be implemented to mitigate the impact on underrepresented groups, particularly Blacks, in STEM fields? What broader economic and policy considerations might influence the trends observed in the engineering workforce for different racial groups?"

In terms of ease of use, participants reported:

"The overall visual setup of the graphs make sense, the data is simple, clear and easy to understand."
"I find the initial graph representations to be excellent. The labeling on the graphs is coherent and provided me with a comprehensive overview at first glance. The filter option was instrumental in refining the information presented in the graph, allowing for a more focused representation and interactivity."

For the second version of the Data Hub, our focus was on the data visualizations, however, participants reflected about the learning curve and highlighted the need for explanations to help new users:

"Engaging in this activity provided me with a deeper comprehension of navigating the system, although the initial experience was admittedly frustrating."

"New users should know where this data came from and the overall purpose of the Data Hub."

In terms of perceived challenges, specific issues emerged related to the meaning of the y-axis value, namely for years during the COVID-19 pandemic. This relates to specific characteristics of the library used to plot and a lack of annotation for certain periods. Desired features pointed out by participants include features data and visualization-related features. Desired features about data include descriptions for engineering data charts and definitions for race and ethnicity. Desired features about the visualization library include label customization, color customization, improved usability for the creation of filters, map visualization, adding callouts to highlight key findings, test statistics, and measures of central tendencies. Although PyGWalker was used due to its similarity to the Tableau user interface, it still has some features missing for the audience considered in this case study. A participant summarized their experience with the prototype by balancing available features with the time required to explore the visualization:

"Success in using the system relies on careful attention to detail and a willingness to invest little time in exploring its capabilities."

In terms of barriers faced, a participant with visual impairment reported issues while interacting with the graphs with the high contrast mode:

"I have a visual impairment to some extent, and I encountered difficulty in discerning details within the graph. I have better visibility with white color shades, and I was specifically looking for an option to switch the color mode, but regrettably, I could not locate any such feature."

Another participant tried accessing the computational notebook from a mobile device and reported not being able to access the visualization due to the improper sizing of visual elements of the computational notebook for mobile devices. Moreover, when trying to export a visualization, a participant discovered an issue related to the environment in which the prototype is running:

"I saw an option to export the results, but it was not working for me (either it was from my side or it has not been implemented yet)."

Further, the implications of the presented case study to Human-Computer Interaction and Data Science point to directions such as the combination of PAR and Data Action methods with computational notebooks as a way to empower people to engage in data analysis conversations, the promotion of meaningful first experiences with data analysis technologies, and the need for data analysis libraries that ease the process of combining multiple publicly available data.

6 CONCLUSION

This case study presented the Data Hub project, which involved a partnership with a non-profit organization supporting Black engineers and aimed at providing an easy to use data analysis prototype so partners could create their own data narratives. Although this case study involved a partnership with a non-profit organization through the Data Hub project, the number of people interacting with the prototypes in focus groups and interviews was small in comparison with more traditional usability studies. However, the reduced number of participants is due to the nature of the project and the PAR foundation of the case study reported. In terms of generalizability, this case study represents an in-depth analysis of how stakeholders from one SJNP interact, reflect, and create their own data narratives. Hence, although consonant with the literature when presenting the benefits of computational notebooks, our work has limited generalization given the scope of the partnership and the very situated nature of a PAR project. As pointed out by [11], PAR projects are inherently embedded in particular concrete contexts and relationships and other forms of mutual learning with vital importance.

Stakeholders from the partnering organization were part of the discussions, provided feedback on the first prototype, and are using the second version of the prototype, now based on a computational notebook. The data analysis provided supported the combination of multiple tables of publicly available data. However, it required considerable effort in understanding data sources and combining tens of tables in both prototypes. Hence, the most important identified need is that *publicly available data sources should provide easier ways for communities to create their own data narratives*. In this case study, participants were able to perform the tasks in the computational notebook provided. Although we obtained valuable feedback for the current version, the identification and analysis of narratives created by them require further investigation, including involving more stakeholders from the non-profit partner organization. These two aspects represent the key next steps for our research.

In light of what is possible to do with Large Language Models (LLMs) today in terms of data analysis, a new path for research resides in data-literacy and AI-literacy for performing data analysis with generative AI. Although limited number of tokens considered for data ingestion (context), lack of explainability, inconsistent outputs, and “hallucinations” are notable challenges to overcome, this is an interesting research path to pursue as well.

Finally, the Data Hub project follows key characteristics of a PAR project, as it aimed at generating knowledge-for-action. To this end, the support from our company on not requiring the creation of a new technology/asset allowed us to truly focus on SJNP needs. In turn we were able to combine theory and practice, which can be summarized by citing a PAR principle that states that *“the research process is as important as the outcomes”* [11].

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